



Assessment of the Effectiveness of Flood Mitigation Measures in Lokogoma, Abuja, Nigeria



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Abstract: Flooding is a recurrent problem in rapidly developing urban districts such as Lokogoma, Abuja, where inadequate drainage, poor land-use planning, and weak infrastructure maintenance increase flood vulnerability. This study assessed the causes, impacts, and effectiveness of existing flood mitigation measures in Lokogoma District, Abuja, Nigeria. A mixed-methods approach was adopted, involving field observation, questionnaire surveys, stakeholder interviews, secondary data review, and geographic information system (GIS)-assisted mapping. A total of 381 questionnaires were administered to residents in flood-prone residential areas, and the data were analyzed using descriptive statistics, mainly frequencies and percentages. The findings show that 79% of respondents had been affected by flooding, while 53.3% reported that flooding occurs frequently. Poor drainage conditions were identified as the leading cause of flooding, accounting for 27.6% of responses, followed by building on waterways and improper planning, each accounting for 22.0%. The major impacts of flooding included disruption of daily activities, economic loss, loss of personal belongings, health issues, property damage, and displacement from homes. Existing mitigation measures included drainage systems, sandbags/barriers, building elevation, flood walls, community education, early warning systems, vegetative barriers, land-use planning, and flood insurance. However, respondents expressed mixed views about their effectiveness, with 44.9% rating the measures as effective or very effective and 39.4% rating them as ineffective or very ineffective. The study concludes that flooding in Lokogoma is driven by inadequate drainage capacity, poor maintenance, improper waste disposal, weak development control, and limited community participation. It recommends upgrading secondary and tertiary drainage systems, improving solid waste management, enforcing land-use and building regulations, conducting regular drainage maintenance, reviewing flood risk assessments, and strengthening public awareness and community participation. These measures can enhance flood resilience and support more sustainable urban development in Lokogoma District.

Keywords: Urban flooding; Flood impacts; Flood mitigation measures; Drainage infrastructure; Land-use planning; Community participation; Urban vulnerability

1 Introduction

Climate change affects human well-being and public health through multiple pathways. By influencing air quality, access to safe drinking water, and the security of human settlements, it can weaken progress in global health improvement and sustainable spatial development [1]. Globally, flooding is one of the most frequent and costly natural hazards [2]. In 2016, for example, a large-scale flood event in the southern United States, Western Europe, several Asian countries, and some parts of Africa caused economic losses of nearly US\$30 billion [3]. Flooding can be triggered by intense precipitation, storm surges, unusually high tides, tsunamis, rapid snow or glacier melt, sea-level rise, and the failure of dams, levees, retention ponds, or other water-control structures [4, 5]. Therefore, flood risk reduction has become closely linked to the Sustainable Development Goals, particularly the need to build resilience and reduce the impacts of climate-related hazards and natural disasters [6].

Nigeria continues to experience severe flood impacts, similar to several other West African countries such as Mali, Senegal, Burkina Faso, and Niger [7]. However, the capacity to respond to and manage flood disasters may differ across countries and regions [2, 6]. Recent studies have identified an increase in the frequency of flood events on a

global scale [8]. This increase has been linked to the growing exposure of people and property, rising sea levels, rapid urban expansion, and more frequent extreme precipitation events [9].

Flooding is no longer a new phenomenon in Nigeria, and its destructive tendencies are sometimes enormous [4, 10]. The country has experienced serious flood-related losses within the West African region. In 2018, the National Emergency Management Agency (NEMA) reported that severe flooding had affected 103 Local Government Areas across 10 states by October 9, with about 1.9 million people impacted. The report further noted that 561,442 people were internally displaced and 351,236 required urgent humanitarian assistance [11].

Flooding is a natural phenomenon that cannot be eliminated, as water must always flow somewhere when it rains [4, 5]. Nevertheless, future flood damage is likely to worsen as populations and economic activities continue to concentrate in flood-prone areas, while climate change increases the likelihood of more intense and extreme weather events [12]. The problems causing excessive seasonal flooding have resulted in great damage to the environment, reduced environmental quality, and threatened the lives of residents [4, 10]. The rate of encroachment, land use change, and the gradual expansion of water bodies will likely increase due to urbanization and unplanned development; therefore, the possibility of flooding will also increase [10, 12].

Lokogoma is a densely populated district in Abuja, Nigeria, characterized by rapid urbanization and extensive residential development [13]. This district has repeatedly experienced severe flooding, despite the implementation of various flood mitigation measures intended to reduce flood impacts [13, 14]. The persistence of these flooding events suggests that the current strategies may be insufficient, poorly maintained, or inadequately integrated with land-use planning and community preparedness [13, 15]. Potential contributing factors include inadequate drainage capacity, blocked or poorly maintained drainage channels, inappropriate development in flood-prone areas, and limited public awareness of flood risk [10, 13–15].

This research aims to investigate the performance and impact of the existing flood mitigation measures and to determine why some non-structural flood mitigation measures fail to address the flooding challenges. By examining these factors in detail, the study aims to uncover the underlying deficiencies and provide a comprehensive understanding of why the current measures fail to mitigate the effects of flooding effectively in the district, as well as the barriers to implementing flood mitigation policies. Against this background, and considering the expected rise in rainfall due to climate change, this study assesses the effectiveness of flood mitigation measures implemented in Lokogoma District, with a view to proposing physical planning recommendations to reduce the effects of flooding in the study area. Specifically, it considers the land use dynamics of the study area, the physical characteristics such as topography, proximity to water bodies, slope, and terrain, as well as the frequency and extent of flood occurrences, in order to propose compatible land use and engineering approaches to flood mitigation in the study area.

2 Literature Review

2.1 Concept of Flooding

Flooding can be defined as the partial or complete inundation of normally dry land by water. It may result from the overflow of rivers, lakes, tidal water, or inland water, as well as from the rapid accumulation of surface runoff [16]. Flooding can also occur when water levels exceed natural or artificial boundaries, such as riverbanks or levees, or when rainfall accumulates on saturated ground [17].

2.2 Types of Flooding

To effectively mitigate the risk of flooding, it is important to understand the various types of floods that can occur. Floods are commonly classified into three main types: fluvial or river floods, pluvial or flash floods, and coastal floods [18]. In Nigeria, flooding has also been discussed in relation to river flooding, urban flooding, and coastal flooding [19].

2.2.1 Fluvial (river) floods

Fluvial floods are mainly caused by prolonged rainfall within or upstream of the affected area. They may also result from rapid snowmelt or the overtopping of flood-control structures such as levees and dikes [18]. When river or stream discharge exceeds channel capacity, water spreads beyond the banks and inundates nearby floodplains or low-lying areas. Under wet conditions, flooding may also occur when a shallow water table rises above the ground surface [17].

2.2.2 Pluvial (flash) floods

Pluvial or flash floods are short-duration events with high peak discharge, usually caused by intense rainfall that exceeds ground infiltration or drainage capacity. They may also occur after dam or levee failure, or the sudden release of water from an ice jam [18]. Rainfall intensity and duration, surface conditions, topography, and basin slope all influence their severity. These floods can damage roads and bridges, destroy houses, endanger lives, and drown livestock [20].

2.2.3 Coastal floods

Coastal flooding occurs when large volumes of seawater are pushed onto normally dry land, primarily as a result of storm surges during severe weather events such as hurricanes or tropical cyclones [18]. Its severity is influenced by strong winds, low atmospheric pressure, high tides, wave action, sea-level change, and coastal topography. Low-lying coastal areas are particularly vulnerable because floodwater can spread farther inland where elevation changes are limited.

2.3 Concept of Flood Mitigation

Mitigation refers to actions taken to reduce disaster-related losses to lives, property, and economic activities. It is usually guided by risk assessment, which helps identify hazards, exposed assets, and appropriate measures for reducing future risk [21]. In the context of flooding, mitigation involves structural and non-structural measures designed to reduce flood impacts on communities, infrastructure, and ecosystems [22]. These measures may include floodwater control, land-use planning, drainage improvement, public awareness, and other policy or management actions aimed at reducing vulnerability. Effective flood mitigation also requires the identification of flood-prone areas and floodplains, followed by the design of appropriate measures based on flood frequency and risk levels [22, 23].

A floodplain is the low-lying land beside a river that may be inundated when river flow rises during periods of heavy rainfall [23]. Mitigation planning focuses on identifying long-term policies and actions that can reduce exposure, limit future losses, and support risk-based decision-making for communities, infrastructure, and economic assets [21]. Therefore, flood mitigation measures serve to reduce the negative impact of flooding and not to completely stop it.

2.3.1 Types of mitigation

Risk avoidance: This strategy involves eliminating or minimizing exposure to flood risk. In flood mitigation, risk avoidance may include avoiding construction in flood-prone areas, restricting development on floodplains, or relocating vulnerable activities away from high-risk zones. Risk avoidance is a primary strategy in disaster risk management, especially in flood-prone areas [22]. Although it may not always be a complete solution, its application during planning and decision-making processes can significantly reduce flood-related losses.

Risk reduction: This approach focuses on lessening the likelihood or severity of flood impacts when risk cannot be completely avoided. Risk reduction seeks to reduce potential damage to people, property, infrastructure, and the environment. It may involve improving preparedness, strengthening local capacity, and reducing vulnerability before flood events occur. Reducing flood risk requires a combination of preventive, protective, and preparedness measures [22].

Risk transfer: Risk transfer refers to shifting part of the financial burden of flood damage to another party. This may be achieved through insurance, compensation schemes, risk-sharing arrangements, or contractual mechanisms. Flood insurance is an important approach for managing the financial consequences of flood events [24]. However, risk transfer does not remove the physical hazard itself, so it should be used together with other forms of mitigation.

2.3.2 Strategies of flood mitigation

Flood mitigation strategies are commonly grouped into structural and non-structural approaches. Structural approaches rely on physical infrastructure to control, divert, store, or regulate floodwater. Common measures include retention ponds, detention basins, self-closing flood barriers, flood walls, sea walls or coastal defenses, diversion channels, dikes, dams, reservoirs, and improved drainage systems [22]. These measures can reduce flood peaks, redirect runoff, and protect settlements and infrastructure from flood damage. Figure 1 illustrates the movement of water through a detention pond-wetlands system [25].

Non-structural approaches focus on reducing flood damage through planning, regulation, awareness, and policy measures rather than relying only on physical infrastructure [22]. These measures include land-use planning, floodplain zoning, building codes, property buyouts, permanent relocation, early warning systems, public awareness, emergency preparedness, and routine drainage maintenance. This approach is particularly important in urban areas where unplanned development, poor waste management, and encroachment on flood-prone land can increase flood risk [19, 22].

2.3.3 Flood mitigation in developed countries using dikes and flood barriers

The construction of dikes and flood barriers is a fundamental component of the Dutch flood defense system [26]. Dikes are earthen structures built along water bodies to prevent flooding, while flood barriers are movable barriers that can be deployed during high water levels to block incoming floodwaters. Figure 2 shows the Maeslant Barrier in Rotterdam [27] and the Oosterscheldekering barrier in the Netherlands [28].

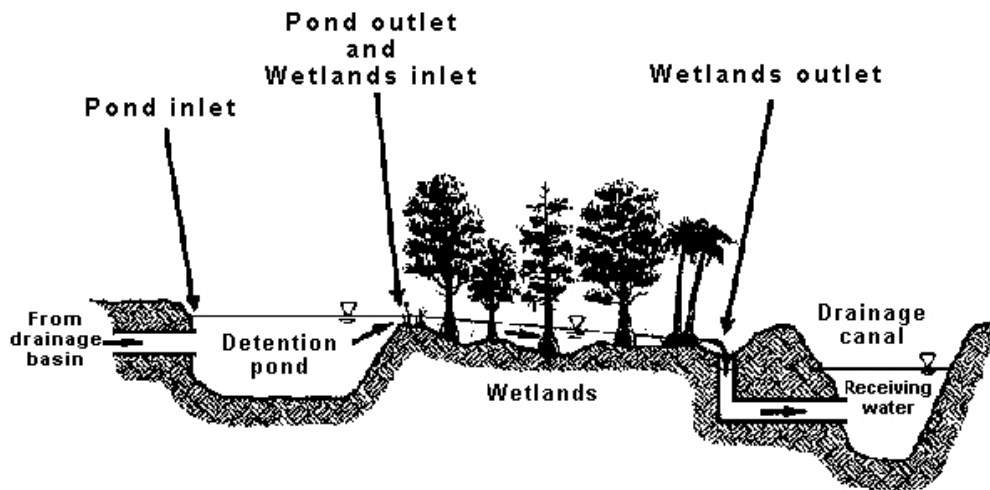


Figure 1. Movement of water through a detention pond-wetlands system

Source: Reproduced from Martin and Smoot [25], U.S. Geological Survey.



(a)



(b)

Figure 2. Maeslant Barrier in Rotterdam (a) and Oosterscheldekering barrier in the Netherlands (b)

Source: Figure 2a adapted from Ref. [27], CC BY-SA 3.0; Figure 2b adapted from Ref. [28], CC BY 3.0.

2.3.4 Flood mitigation in Nigeria using construction of dams and reservoirs

Flood mitigation strategies in developing countries must often be tailored to the specific challenges and resources available in these regions [22]. In Nigeria, flood mitigation strategies encompass various approaches, including structural and non-structural measures [19]. Building dams and reservoirs helps regulate water flow, reduce flood peaks, and provide water storage for irrigation and other purposes. For example, the Kainji Dam on the Niger River serves flood protection purposes along with hydroelectric power generation and navigation improvement [29].

3 Methodology

This study adopted a mixed-methods approach involving field observation, questionnaire surveys, stakeholder interviews, secondary data review, and geographic information system (GIS)-assisted mapping. These methods were used to obtain the information needed to describe the existing situation and identify requirements for improvement.

3.1 The Study Area

Lokogoma developed as part of the broader expansion of Abuja's residential districts following the relocation of Nigeria's federal capital from Lagos to Abuja in 1991 [30, 31]. Lokogoma, a district in southwestern Abuja, Nigeria's capital, has undergone significant transformation over the past few decades. Originally a sparsely populated area with limited infrastructure, Lokogoma rapidly evolved into a residential hub as Abuja expanded to accommodate civil servants, business professionals, and other residents migrating to the new capital region. The original residents were the indigenous Gwari people, who primarily engaged in farming and other agricultural activities [31].

It is situated within the Abuja Municipal Area Council (AMAC) and is known for its rapidly developing residential areas. Lokogoma is located at approximately 8.9750° N latitude and 7.4720° E longitude. It is bounded by Gaduwa to the north, Wumba to the east, Kabusa to the south, and Galadimawa to the west. Figure 3 shows the locational map of the study area.

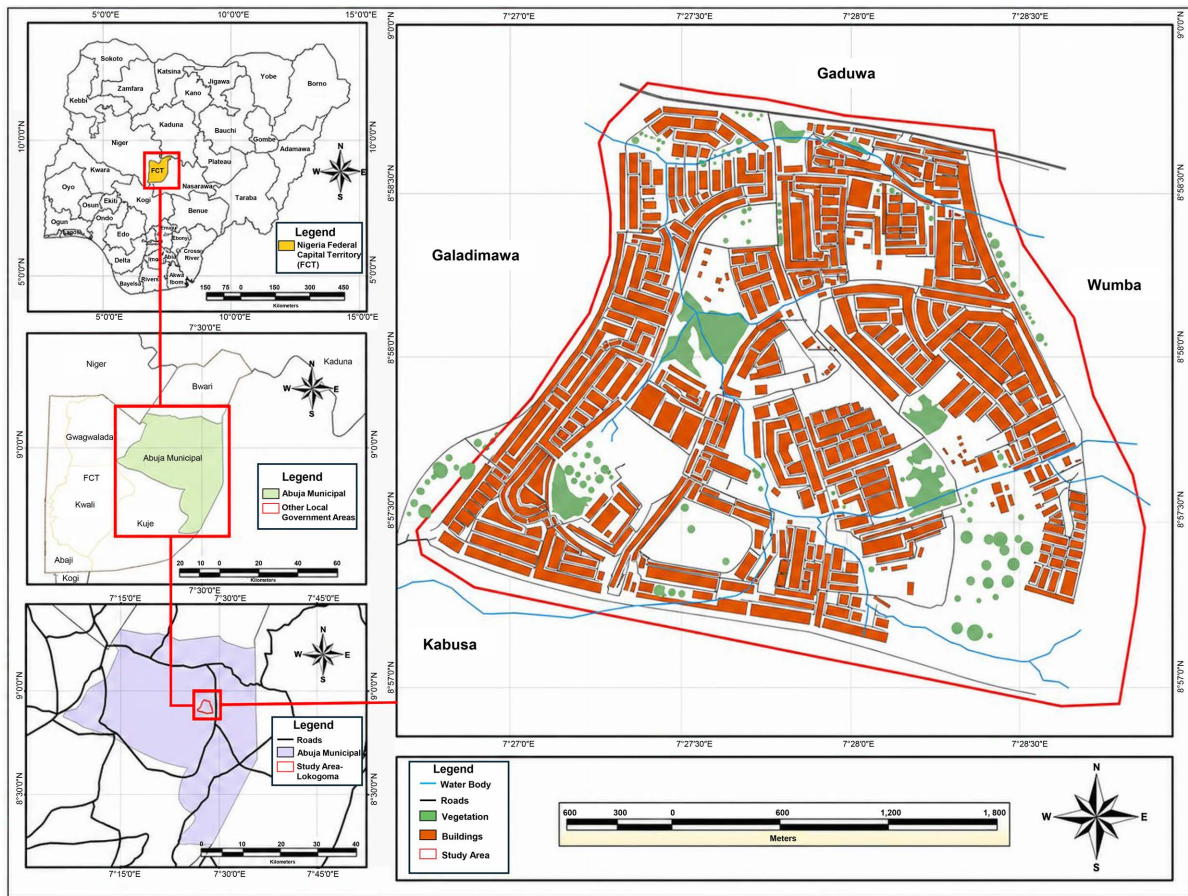


Figure 3. Location map of Lokogoma District within Abuja, Nigeria

Source: Author's own mapping, 2025.

3.2 Data Collection

3.2.1 Primary data

Primary data were collected through field observation, questionnaire administration, and semi-structured interviews. The field activities focused on the nature of the floodplains, the occurrence of flooding, flood-related damage, the condition of drainage systems, and existing flood mitigation methods.

(1) Reconnaissance survey: A pre-field survey was carried out to obtain a clear understanding of the study area. Spatial information, including satellite imagery from Google Earth and other relevant mapping data, was used to prepare the location map of the study area. ArcMap was used to digitize, label, and present the spatial features shown in the map [32, 33].

(2) Field survey: A field survey was carried out to obtain a general overview of the study area and to observe drainage conditions, flood-prone locations, land use patterns, and existing flood mitigation measures.

(3) Questionnaires: A total of 381 questionnaires were administered to residents of the study area using a systematic random sampling technique at a five-household interval. The questionnaire survey was designed to capture residents' experiences with flooding, including the frequency and depth of flood events, the extent of damage incurred, and their perception of the effectiveness of current mitigation measures.

(4) Interviews: Semi-structured interviews were conducted with key stakeholders involved in flood risk management in the study area. These stakeholders included local government officials responsible for infrastructure development and maintenance, town planning authorities who oversee land-use regulations, and representatives of disaster management agencies who play a critical role in emergency preparedness and response. The interviews aimed to understand the perspectives of these stakeholders on flood risks, existing mitigation strategies, and potential areas for improvement.

3.2.2 Secondary data

Secondary data were collected through a review of academic literature, technical reports, and other relevant documents on urban flooding and flood mitigation. The reviewed materials covered structural and non-structural flood mitigation measures, as well as case studies of flood management practices implemented in other cities. This review provided background information on flooding, flood causes, and mitigation approaches relevant to the study. Spatial information, including satellite imagery and other relevant mapping data, was also used to prepare the location map of the study area.

3.3 Data Analysis

The questionnaire data were analyzed using descriptive statistics, mainly frequencies and percentages. The results were presented in tables, charts, and figures to show respondents' experiences with flooding, perceived causes of flooding, types of damage, existing mitigation measures, community participation, and the perceived effectiveness of current flood mitigation measures. Interview responses and field observations were used to support the interpretation of the questionnaire results. ArcMap was used only to prepare the location map of the study area, as shown in Figure 3.

4 Results and Discussion

4.1 Drainage Characteristics

Studying drainage characteristics is key to understanding how water flows and is managed in an area. This includes looking at different types of drainage systems, such as primary drains, secondary drains, and tertiary drains. The sizes of these drains, whether large enough to handle stormwater or too small, affect how well they prevent flooding. In Lokogoma, analyzing these systems helps identify issues such as blocked or undersized drains, leading to recommendations for better maintenance or resizing to reduce flood risks.

Table 1 shows the current dimensions of drainage systems in the study area. The primary drainage, with a width of 3 m and a depth of 2 m, appears to have a relatively larger capacity for managing stormwater compared with the secondary and tertiary drainage systems. The secondary drainage, at 1.05 m wide and 1 m deep, is relatively narrower, which may limit its efficiency. The tertiary drainage, with a width of 0.5 m and a depth of 0.35 m, may be insufficient for handling localized runoff effectively.

Table 1. Drainage types and sizes

Types	Width (m)	Depth (m)
Primary	3	2
Secondary	1.05	1.00
Tertiary	0.50	0.35

Source: Field measurement, 2025.

The drainage data suggest that flood problems in Lokogoma are related not only to the existence of drainage channels but also to their carrying capacity and maintenance condition. While the primary drainage system appears relatively larger, the smaller dimensions of secondary and tertiary drains may limit the movement of runoff within residential areas. Previous studies have shown that inadequate drainage infrastructure, poor urban planning, and weak maintenance of flood-control systems can increase flood risk in rapidly growing urban areas [10, 22]. Thus, improving local drainage channels is as important as maintaining the major drainage infrastructure.

4.2 Respondents' Experience with Flooding

The survey results show that most respondents had experienced flooding in the study area. Based on Figure 4, most respondents, 79%, reported that they had been affected by flooding, while 21% stated that they had not experienced flooding. This high proportion of affected respondents indicates that flooding is a widespread issue within the community, emphasizing the need for effective mitigation measures and long-term solutions.

The survey also examined how often flooding occurs in the study area. As shown in Table 2, a majority, 53.3%, reported that floods occur frequently, multiple times a year, indicating that flooding is a persistent and regular issue in the area. In addition, 45.4% responded that floods occur occasionally, further emphasizing the recurring nature of the problem. Only a small minority, 1.3%, noted that floods happen rarely, occurring once in several years. This distribution highlights the widespread and frequent occurrence of floods, suggesting that flooding is a significant and ongoing concern in the community. The data underscore the urgency for effective flood management and mitigation strategies.

These findings suggest that flooding in Lokogoma is a recurring urban planning and infrastructure problem rather than a temporary seasonal issue. This pattern is consistent with previous studies showing that flood frequency and duration have increased globally, while urban expansion, exposure growth, and inadequate infrastructure can increase

flood vulnerability in cities [8, 9, 22]. For Lokogoma, the repeated exposure of residents to flooding shows that mitigation efforts should focus more on reducing everyday vulnerability, not only on responding after flood events occur.

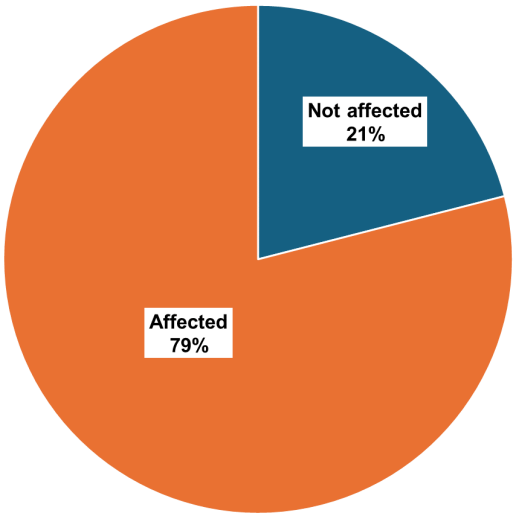


Figure 4. Percentage of respondents affected by flooding
Source: Questionnaire survey, 2025.

Table 2. Frequency of flooding

How Frequently do Flood Occur?	No. of Respondents	Percentage (%)
Rarely	5	1.3
Occasionally	173	45.4
Frequently	203	53.3
Total	381	100.0

Source: Questionnaire survey, 2025.

4.3 Causes of Flooding

Table 3 presents the respondents’ views on the causes of flooding, providing insight into the community’s understanding of flood risks. The most frequently mentioned cause was poor drainage conditions, accounting for 27.6% of respondents, suggesting that inadequate drainage systems are a significant contributor to flooding. Both building on waterways and improper planning accounted for 22.0% of respondents, indicating that human activity and a lack of foresight in urban development are also key factors. Improper waste disposal was noted by 13.9% of respondents, reflecting the role of waste management in exacerbating flooding issues. Additionally, 14.4% of respondents attributed flooding to choked drains, emphasizing the impact of blocked or neglected drainage systems. The responses suggest that a combination of poor infrastructure, improper urban development, and waste management issues is among the main causes of flooding. Figure 5 shows blocked drainage channels observed during the field survey, supporting the respondents’ view that poor drainage conditions contribute to flooding in the study area.

Table 3. Causes of flooding

Causes of Flooding	No. of Respondents	Percentage (%)
Building on waterways	84	22.0
Poor drainage conditions	105	27.6
Improper waste disposal	53	13.9
Improper planning	84	22.0
Choked drains	55	14.4
Total	381	100.0

Source: Questionnaire survey, 2025. Note: Percentages may not sum to 100.0 due to rounding.



Figure 5. Blocked drainage channels in the study area

Source: Author's field photograph, 2025.

The reported causes show that flooding in the study area is largely influenced by human activities and infrastructure-related problems. Poor drainage conditions, building on waterways, improper planning, and waste disposal practices all point to weaknesses in urban management. The blocked drainage channels observed during the field survey further support this pattern. Similar findings have linked urban flooding to inadequate drainage infrastructure, poor land-use control, blocked drains, and weak maintenance of drainage systems [10, 19, 22]. Therefore, addressing the flooding problem requires attention to both physical infrastructure and the regulation of urban development.

4.4 Damage Caused by Flooding

The types of damage caused by flooding were identified by the respondents, as shown in Table 4. The most cited impact was the disruption of daily activities, accounting for 18.9%, which shows how floods interfere with regular routines and productivity in the community. Economic loss was also notable at 17.3%, followed by the loss of personal belongings at 16.8%, indicating that floods not only affect people's day-to-day lives but also result in financial hardship. Health issues represented 16.3% of responses, reflecting the public health risks associated with flooding, such as waterborne diseases. Property damage was recorded at 15.7%, while displacement from homes accounted for 15%, further emphasizing the severe personal and structural toll that floods take on the community. This diverse range of flood impacts illustrates the widespread consequences of flooding, affecting everything from livelihoods to health, and highlights the need for comprehensive flood response and recovery measures.

Table 4. Damages caused by flooding

Types of Damage Caused by Flooding	No. of Respondents	Percentage (%)
Property damage	60	15.7
Loss of personal belongings	64	16.8
Displacement from homes	57	15.0
Health issues	62	16.3
Disruption of daily activities	72	18.9
Economic loss	66	17.3
Total	381	100

Source: Questionnaire survey, 2025.

The range of damages reported by respondents shows that flooding affects more than buildings and physical infrastructure. Disruption of daily activities, economic loss, health issues, and loss of belongings indicate that flooding directly affects household welfare and community stability. These effects are consistent with earlier reports showing that floods can disrupt livelihoods, damage property, increase health risks, and displace vulnerable populations [7], [22]. The results therefore suggest that flood management in Lokogoma should consider social and economic impacts alongside physical flood control.

4.5 Existing Flood Mitigation Measures

Based on the recurrent flooding problems identified in the study area, respondents reported the use of both structural and non-structural measures to mitigate flood impacts. These measures are discussed in this section.

4.5.1 Structural mitigation measures

Table 5 outlines the structural measures used to mitigate flooding. A total of 32.8% of respondents reported the use of drainage systems, reflecting the importance of proper infrastructure in flood management. Sandbags/barriers ranked second, accounting for 28.9% of respondents, indicating a reliance on temporary solutions during flood events. Building elevation, adopted by 20.7%, represents a long-term approach to protecting homes from floodwater. In addition, 17.6% of respondents indicated the use of flood walls as a physical barrier against flooding. These responses highlight both temporary and permanent solutions, with drainage systems and sandbags being the most widely used strategies.

Table 5. Structural mitigation measures used by respondents

Structural Mitigation Measure	No. of Respondents	Percentage (%)
Drainage systems	125	32.8
Elevation of buildings	79	20.7
Flood walls	67	17.6
Use of sandbags/barriers	110	28.9
Total	381	100.0

Source: Questionnaire survey, 2025.

The dominance of drainage systems among the structural measures shows that residents associate flood control mainly with physical infrastructure. However, the relatively high use of sandbags/barriers suggests that many households still depend on temporary responses during flood events. Structural measures such as drains, flood walls, detention basins, and other engineering works can reduce flood impacts, but their effectiveness depends on proper design, maintenance, and coordination with non-structural measures [22]. This may explain why residents continue to rely on short-term protective measures despite the presence of drainage infrastructure.

4.5.2 Non-structural mitigation measures

Respondents reported the use of several non-structural and nature-based measures to address flooding, as presented in Table 6. Community education was the most reported measure, accounting for 27.8% of respondents. Vegetative barriers accounted for 23.4%, suggesting that some residents recognize the role of natural features in reducing runoff and flood impacts. Early warning systems were reported by 22.6% of respondents, indicating the importance of timely information before flood events. Land-use planning accounted for 18.4%, reflecting the role of development control in reducing flood exposure. Finally, flood insurance was reported by 7.9% of respondents, suggesting limited attention to financial preparedness and risk transfer.

Table 6. Non-structural and nature-based mitigation measures used by respondents

Mitigation Measure	No. of Respondents	Percentage (%)
Community education	106	27.8
Early warning systems	86	22.6
Vegetative barriers	89	23.4
Land-use planning	70	18.4
Flood insurance	30	7.9
Total	381	100.0

Source: Questionnaire survey, 2025. Note: Percentages may not sum to 100.0 due to rounding.

The responses show that residents recognize the value of non-structural measures, especially community education, early warning systems, and land-use planning. However, the low use of flood insurance indicates limited attention to financial preparedness and risk transfer. Previous studies have emphasized that non-structural measures, including land-use planning, public awareness, early warning, and insurance, can reduce flood vulnerability when they are properly implemented [21, 22, 24]. In Lokogoma, these measures appear to exist, but their use remains uneven across the community.

4.6 Community Participation in Flood Mitigation

Figure 6 shows the level of community participation in efforts to manage or mitigate floods. A significant majority, 60%, of respondents reported being involved in such efforts, indicating strong community engagement in addressing the flooding issue. However, 40% of respondents stated that they did not participate, highlighting a substantial percentage of the population that is either disengaged or unable to contribute to flood management initiatives. This suggests the need to improve awareness and mobilize more community members to participate in flood mitigation activities. Encouraging greater involvement could enhance the overall effectiveness of flood management strategies and strengthen collective action in addressing local flood risks.

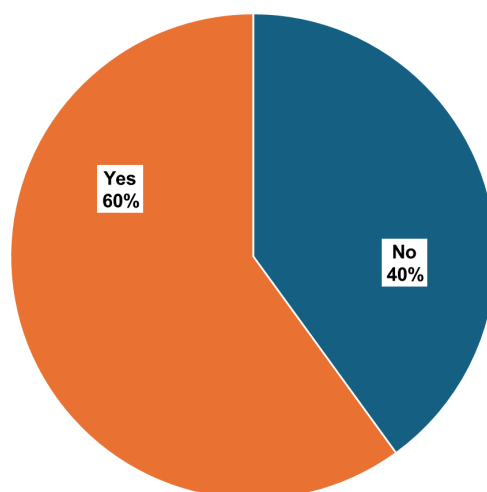


Figure 6. Participation in community-driven flood mitigation projects

Source: Questionnaire survey, 2025.

Although many respondents reported participating in flood mitigation activities, the proportion of non-participants remains considerable. This matters because local flood control often depends on collective actions, such as keeping drains clear, disposing of waste properly, responding to warnings, and reporting unsafe development practices. Previous studies have emphasized that community participation and local preparedness can support the implementation and maintenance of flood mitigation measures [21, 22]. Strengthening participation in Lokogoma could therefore improve the practical effectiveness of both structural and non-structural measures.

4.7 Effectiveness of Current Flood Mitigation Measures

Respondents expressed varying opinions regarding the effectiveness of the flood mitigation measures currently in place, as presented in Table 7. A total of 20.5% rated the measures as very effective, while 24.4% considered them effective. The combined total of 44.9% indicates that nearly half of the respondents had a positive view of the existing flood control efforts. In contrast, 25.5% believed the measures were ineffective, 13.9% rated them as very ineffective, and 15.7% remained neutral, suggesting some uncertainty or mixed experiences. This distribution reflects a divided opinion on the effectiveness of flood mitigation, pointing to the need to reassess and improve existing measures to address the flooding problem.

Table 7. Effectiveness of current flood mitigation measures

How Effective are the Current Flood Mitigation Measures?	No. of Respondents	Percentage (%)
Very effective	78	20.5
Effective	93	24.4
Neutral	60	15.7
Ineffective	97	25.5
Very ineffective	53	13.9
Total	381	100.0

Source: Questionnaire survey, 2025.

The mixed perception of effectiveness suggests that existing mitigation measures have achieved some positive results but are not yet sufficient for many residents. The share of respondents who rated the measures as ineffective or very ineffective points to possible gaps in maintenance, enforcement, coordination, and public participation. Previous

studies have shown that flood mitigation measures may be weakened when drainage infrastructure is poorly maintained, land-use controls are inadequate, or community participation is limited [19, 22]. A more coordinated approach is therefore needed, combining drainage improvement, development control, waste management, and community involvement.

5 Summary and Recommendations

5.1 Summary of Findings

The major findings of the study are summarized as follows:

- Insufficient and poorly maintained drainage systems were identified as the major factors contributing to flooding in the study area.
- Poor urban planning and development on waterways or flood-prone areas were found to increase flood risk in some parts of Lokogoma.
- A gap exists between flood mitigation policies and their effective implementation, particularly in relation to development control, drainage maintenance, and waste management.
- Some houses and other structures have been developed on natural floodplain areas that are inherently susceptible to flooding.
- In some parts of Lokogoma, residents dump waste in constructed drainage channels, which leads to clogging and prevents the free flow of water.
- Public awareness and community participation in flood risk reduction remain insufficient and need to be strengthened.

5.2 Recommendations

Based on the findings, the following recommendations are proposed to address the flooding problem and improve flood mitigation practices in the study area.

1. **Drainage Expansion and Capacity Improvement:** Existing drainage systems should be upgraded by increasing their dimensions and improving their flow capacity to handle heavy rainfall and stormwater. Priority should be given to secondary and tertiary drains, which are more directly connected to residential areas and may be insufficient for localized runoff.
2. **Development of Designated Waste Disposal Sites:** Improper disposal of waste, especially into drainage systems and waterways, is a significant contributor to flooding in Lokogoma. Well-planned waste disposal sites should be provided to reduce indiscriminate dumping of refuse and prevent drainage blockage during the rainy season.
3. **Regular Waste Collection Schedule:** A consistent waste collection schedule should be established for residential and commercial areas. Waste should be collected frequently enough to prevent the accumulation of refuse in drainage channels and open spaces.
4. **Drainage System Cleaning and Maintenance:** Routine cleaning of drains and waterways should be scheduled to remove accumulated waste and debris that obstruct water flow. Waste management teams and relevant local authorities should periodically inspect and clean drainage systems, especially before and after the rainy season.
5. **Flood Risk Assessment Reviews:** Flood risk assessments for buildings should be periodically reviewed, particularly after major weather events or new urban developments. Environmental and urban planning experts should collaborate with estate managers and local authorities to reassess flood-prone areas and update planning controls where necessary.
6. **Compliance Monitoring with Building Codes:** New constructions and renovations should comply with updated building codes designed for flood resilience. Urban planning authorities should strengthen the enforcement of physical planning standards through building permits, regular site inspections, and penalties for non-compliance. Buildings found to be non-compliant should be required to make necessary adjustments.
7. **Public Awareness and Community Engagement:** Public awareness campaigns should be organized to educate residents on flood risks, proper waste disposal, drainage maintenance, and the importance of avoiding construction on flood-prone areas. Greater community participation will help improve the effectiveness of both structural and non-structural flood mitigation measures.

5.3 Conclusion

The study concludes that inadequate drainage, poor urban planning, weak regulatory enforcement, and improper waste disposal contribute to flooding in Lokogoma. Limited public awareness and uneven community participation further reduce the effectiveness of existing flood mitigation efforts. Although some mitigation measures have been put in place over the years, they have not been fully effective in managing floods in the study area due to poor maintenance, inadequate monitoring, and the lack of coordinated structural and non-structural mitigation strategies. Effective flood mitigation will require infrastructure improvements, stricter regulatory enforcement, regular maintenance, improved

waste management, and active community engagement. Addressing these issues can enhance resilience and promote sustainable development in the area.

Informed Consent Statement

Informed consent was obtained from all participants involved in the questionnaire survey and interviews before data collection. Participation was voluntary, and respondents were informed that their responses would be used only for research purposes.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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